

PRISCILA DE AZEVEDO DRUMMOND

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RESEARCH INTERESTS

Coordinated decision-making across organizations · Prescriptive analytics for multi-agency intervention · Resource allocation in constrained service and supply systems · Supply chain compliance and labor risk · Measurement design for contested and multi-dimensional outcomes · Responsible analytics and the consequences of optimized decisions · Analytics and operations pedagogy

Methods: network optimization and interdiction · integer and mixed-integer programming · Bayesian decision analysis and cost-benefit modeling · multilevel and hierarchical regression · data envelopment analysis · expert elicitation · systematic and scoping review · qualitative and participatory methods

EDUCATION

Northeastern University, Boston, MA

Ph.D., Industrial Engineering, expected December 2026 (defense scheduled 2 December 2026)

Dissertation: *Collaborating Agents and Network Interdiction Models*

Advisor: Dr. Kayse Lee Maass. Committee: Dr. Shahin Shahrampour, Dr. Gabriela Gongora Svartzman

Northeastern University, Boston, MA

Certificate, Post-secondary Teaching, Spring 2026

Capstone: Course design for *System Engineering in Public Programs* (Graduate level) and *Deterministic Operations Research* (Graduate level)

University of Texas Rio Grande Valley, Edinburg, TX

M.S., Engineering Management, 2021

Thesis: *A Framework to Identify Human Trafficking at the U.S. Border* (Advisor: Dr. Hiram Moya)

Universidade Federal do Rio Grande do Sul, Porto Alegre, Brazil

M.S., Business Administration, 2019

Thesis: *Relative Efficiency Analysis of the Women's Crisis Center Network in Rio Grande do Sul* (Advisor: Dr. Denise Lindstrom)

Universidade do Estado do Rio de Janeiro, Rio de Janeiro, Brazil

M.P.S., Finance and Planning, 2016

Project: *Financial and Sporting Efficiency of Brazilian Soccer Clubs* (Advisor: Dr. Francisco Pessanha)

Universidade Federal do Rio de Janeiro, Rio de Janeiro, Brazil

B.A., Industrial Design - Product Design, 2011

Project: *Trebuchet: A Multidisciplinary Pedagogical Game* (Advisor: Dr. Roosevelt Telles)

REFEREED JOURNAL ARTICLES

Jafari, A., Bhimani, S., **Drummond, P. A.**, Nishigaya, D., Ding, A. A., Farrell, A., & Maass, K. L. (2024). Enhancing detection of labor violations in the agricultural sector: A multilevel generalized linear regression model of H-2A violation counts. *PLOS ONE*, 19(5), e0302960. <https://doi.org/10.1371/journal.pone.0302960>

MANUSCRIPTS UNDER REVIEW

Drummond, P. A., & Maass, K. L. Multiple non-independent interventions on the same network element: A scoping review. Under review at *Networks* (submitted September 2026).

Jafari, A., Bhimani, S., **Drummond, P. A.**, Farrell, A., & Maass, K. L. Network interdiction to improve labor trafficking detection in the U.S. agricultural sector. Revise and resubmit at *Decision Sciences*.

REFEREED CONFERENCE PROCEEDINGS

Drummond, P. A., & Moya, H. (2020). U.S. ports of entry's decision model with an anti-human trafficking focus. *Decision Sciences Institute Proceedings*. ISSN 2471-884X

WORKING PAPERS

Drummond, P. A., & Maass, K. L. Collaboration in anti-trafficking efforts: A network interdiction problem with multiple dependent attacks. Status: results analysis. Target: *Production and Operations Management* (September 2026). **[Job Market Paper]**

Drummond, P. A., & Maass, K. L. Optimal bounds and complexity for network interdiction with dependent multiple interventions. Status: results analysis. Target: *IIE Transactions*.

Drummond, P. A., & Maass, K. L. Algorithms for network interdiction problems with non-convex functional attacks. Status: model and algorithm ready; coding. Target: *EJOR*.

RESEARCH IN PROGRESS

Drummond, P. A., & Maass, K. L. Unintended consequences of enforcement in labor trafficking networks: geographic displacement, operational fragmentation, and compliance switching. Status: ideation and literature review stage; resuming after degree completion.

Drummond, P. A. An auto-ethnographic account of a student's experience as a teaching fellow in industrial engineering. Status: data collection, cleaning, and initial literature review ready; resuming after degree completion.

CONFERENCE PRESENTATIONS

Drummond, P. A.*, & Maass, K. L. (2026, November). Network interdiction model with multiple dependent attacks, job market talk. *INFORMS Annual Meeting*, San Francisco, CA.

Drummond, P. A.*, & Maass, K. L. (2025, October). Collaboration in anti-trafficking efforts: A network interdiction problem. *INFORMS Annual Meeting*, Atlanta, GA.

Drummond, P. A.*, Jafari, A., & Maass, K. L. (2024, October). Modeling strategies to disrupt labor trafficking. *INFORMS Annual Meeting*, Seattle, WA.

Drummond, P. A.*, Jafari, A., & Maass, K. L. (2023, October). Labor trafficking disruption with collaborating agents. *INFORMS Annual Meeting*, Phoenix, AZ.

Jafari, A.*, **Drummond, P. A.**, Bhimani, S., Farrell, A., & Maass, K. L. (2023, October). Breaking the chains: An innovative network interdiction approach to combat labor trafficking in agricultural supply chains. *INFORMS Annual Meeting*, Phoenix, AZ.

Drummond, P. A., Jafari, A., Childress, C., & Maass, K. L. (2022, November). Leveraging public data from illicit activities in operations research. *Decision Sciences Institute Annual Conference*.

Drummond, P. A., & Maass, K. L. (2022, October). International migration and human trafficking networks. *INFORMS Annual Meeting*.

Jafari, A.*, **Drummond, P. A.**, Nishigaya, D., Bhimani, S., Farrell, A., & Maass, K. L. (2022, October). Labor violation trends in H-2A nonimmigrant agricultural workers: Improving inspection strategies for detecting labor violations. *Society for Risk Analysis Annual Meeting*.

Gurrola, N.*, Hallas, K.*, **Drummond, P. A.***, Rodriguez, A., & Cano, S*. (2021, April). Understanding female STEM identity at UTRGV through narrative inquiry. *Engaged Scholar Symposium*, University of Texas Rio Grande Valley, Edinburg, TX. Recording: <https://www.youtube.com/watch?v=8gJWmTL9PK0>

Drummond, P. A.*, & Moya, H. (2021, April). Optimization of port-of-entry operation in the U.S.: An anti-human trafficking focus. *5th Coastal Bend Mathematics and Statistics Conference*, Texas A&M University, College Station, TX.

Drummond, P. A.*, & Moya, H. (2020, November). U.S. ports of entry's decision model with an anti-human trafficking focus. *Decision Sciences Institute Annual Conference*, Dallas, TX (virtual).

Drummond, P. A.*, & Moya, H. (2020, November). A framework to identify human trafficking at the U.S. border. *IIE Annual Conference and Expo*, New Orleans, LA (virtual).

Drummond, P. A.*, Rahal, J.*, & Amorim, F. (2008, June). Analysis of different educational strategies using work-based pedagogy at the middle school Colegio de Pescadores, Macae, RJ. *5o Congresso de Extensao da UFRJ*, Rio de Janeiro, Brazil.

POSTER PRESENTATIONS

Drummond, P. A.*, & Maass, K. L. (2024, March). When to collaborate? Labor trafficking interdiction with multiple agents. *RISE*, Northeastern University, Boston, MA.

Drummond, P. A.*, & Maass, K. L. (2023, November). Labor trafficking disruption with collaborating agencies. *MIE Research Expo*, Northeastern University, Boston, MA.

Drummond, P. A.*, & Moya, H. (2021, February). A decision support tool at the U.S. port of entry with a human trafficking focus. *Engineering Week*, University of Texas Rio Grande Valley, Edinburg, TX.

Drummond, P. A.*, & Moya, H. (2020, February). A framework to identify human trafficking at the U.S. border. *Engineering Week*, University of Texas Rio Grande Valley, Edinburg, TX.

Drummond, P. A.*, & Moya, H. (2019, November). A framework to identify human trafficking at the U.S. border. *Graduate Research Fair*, University of Texas Rio Grande Valley, Edinburg, TX.

RESEARCH EXPERIENCE

ORSJ, Northeastern University, Boston, MA

Doctoral Researcher, 2021 to present

Developing network optimization models for multi-agency coordination against labor trafficking, including a new formulation for dependent interventions on a shared network. Cleaned and analyzed three public datasets (500,000+ records) in Python to study labor violation patterns and inspection strategy. Contributed to a multilevel regression study of H-2A violations with an interdisciplinary team spanning criminal justice, social service, engineering, and business. Elicited intervention probabilities from anti-trafficking specialists for use as model parameters. Translated qualitative data to useful forms for Operations Research models (federal cases). Partners: U.S. Department of Homeland Security CINA Center; criminal justice researchers at Northeastern, legal advocacy organization.

University of Texas Rio Grande Valley, Edinburg, TX

Lead Researcher, Female Students' Perception of STEM Identity, October 2020 to June 2021

Originated the study and secured IRB approval; Led a five-member undergraduate research team. Recruited and interviewed 11 participants using a semi-structured protocol. Analyzed first-phase data using narrative inquiry with a priori and emergent coding and member checking, applying narrative inquiry and critical-consciousness frameworks. Disseminated findings at the Engaged Scholar Symposium and the RGV STEM Education Consortium (2021); co-authors continued the study after 2021, and it was published as Hallas & Chapman (2024), RGV STEM Ed Conference Proceedings, <https://doi.org/10.51734/yggj4440>.

Graduate Researcher, 2019 to 2021

Built a Bayesian decision model integrating cost-benefit analysis and statistical risk predictors to determine screening thresholds for trafficking detection at U.S. ports of entry.

Universidade Federal do Rio Grande do Sul, Porto Alegre, Brazil

Graduate Researcher, 2017 to 2019

Partnered with the Rio Grande do Sul Public Security State Department to evaluate resource allocation across the network of services for survivors of domestic violence. Mapped multi-service provider flows, survivor pathways, and information pathways through the system. Conducted data envelopment analysis on crime and social service data across 450+ municipalities using R, combined with structured expert validation. Found that the range of services available constrained effectiveness more than total investment.

Universidade do Estado do Rio de Janeiro, Rio de Janeiro, Brazil

Graduate Researcher, 2015 to 2016

Applied data envelopment analysis to 42 Brazilian soccer clubs, benchmarking financial inputs against sporting performance to identify inefficiency in spending and investment.

Universidade Federal do Rio de Janeiro, Rio de Janeiro, Brazil

Research Coordinator, STEM Pedagogy Assessment, 2006 to 2009

Coordinated a faculty group assessing the outcomes and challenges of a learning-by-working pedagogical model at a project-based public middle school serving a gang-controlled community in Macae, RJ. Qualitative design; results presented at a university extension conference in 2008.

GRANTS AND COMPETITIVE FUNDING

- Dissertation Completion Fellowship, Northeastern University, 2026
- Teaching Fellowship, Northeastern University, Fall 2025 - Spring 2026 (Yearly student awarded fellowship)
- Dean's Chair Fellowship, Northeastern University, 2021
- Presidential Graduate Research Award, UTRGV, 2019
- CNPq Research Award, 2017 (Brazilian competitive research grant and scholarship)
- Travel Funds, PhD Network, Northeastern University, 2022, 2023, 2024, 2025

HONORS AND AWARDS

- College of Engineering Outstanding Graduate Teaching Award, Northeastern University, 2026 (college-level award)
- Tau Beta Pi, Engineering Honor Society, member since 2021
- Outstanding Student Award for Academic Achievement, UTRGV, 2020

TEACHING EXPERIENCE

Instructor of Record

Northeastern University, Boston, MA, Spring 2026

OR 6205: *Deterministic Operations Research* (graduate, 4 credits, 6 enrolled), Spring 2026

Full responsibility for course design, instructional materials, in-class active learning exercises, assessment design, lecturing, grading and feedback, and project mentoring.

Student evaluations rated above departmental means for instructor enthusiasm, feedback sufficiency, an inclusive learning environment, and organization of course materials, and above departmental means for the online learning environment on organization of weekly materials, connection with the instructor, and community among classmates; full TRACE report in the teaching portfolio: <https://pridrum.github.io/teaching/>

Graduate Teaching Assistant

Northeastern University, Boston, MA, 2023 to 2026

- IE 7200: *Supply Chain Engineering* (graduate), Summer II 2026, Prof. A. Guschin, 11 students
- IE 6200: *Engineering Probability and Statistics* (graduate), Summer I 2026, Prof. Doug Schmucker, revision materials
Analyzed the materials and assignments against AI usage; reviewed material from student perspective; created answers with different prompts using AI to test robustness of assignments to use of AI; Weekly meetings with professor, and feedback report for all course materials.
- IE 5617: *Lean Concepts and Applications* (undergraduate and graduate)
 - Summer II 2026, Prof. A. Guschin, 22 students;
 - Summer I 2025, Prof. H. McManus, 20 students;
 - Summer II 2023, Prof. A. Guschin, 46 students;
 - Summer I 2023, Prof. H. McManus, 45 students

Taught the Process Control and Six Sigma Basics module and the Quality Tools and Topics module independently (Summer I 2025). In the end-of-term student ranking of all 17 class sessions, these placed 3rd and 7th, at 4.81 and 4.67 on a 5-point scale, above the instructor's own scores for the same sessions.

- IE 4516: *Quality Assurance* (undergraduate), Fall 2023, Prof. H. McManus - course design

Led a complete redesign of the course teaching materials, restructuring them around experimental, simulation-based, and project-based learning, and taught several of the resulting modules.

Across all Northeastern appointments: developed supporting instructional materials and interactive exercises, held office hours, supported weekly recitations, and graded homework and projects.

University of Texas Rio Grande Valley, Edinburg, TX, 2019 to 2021

- *Engineering Management* (undergraduate), Summer 2021, Prof. H. Moya, 20 students
- *Manufacturing Production and Control* (undergraduate), Fall 2020, Prof. H. Moya, 18 students
- *Operations Research* (undergraduate), Spring 2020 and Spring 2021, Prof. H. Moya, 21 students

Created targeted tutorial sessions and instructional videos during the pandemic transition; implemented personalized feedback systems and supported weekly recitations.

Universidade Federal do Rio Grande do Sul, Porto Alegre, Brazil, 2017 to 2019

- *Manufacturing Production and Control*, Fall 2017, Spring 2018, Spring 2019, Prof. D. Lindstrom, 30+ students per term
- Delivered bi-weekly research seminars and provided project feedback for industry-partnered student work.

Prior Teaching and Curriculum Leadership

Universidade Federal do Rio de Janeiro, Rio de Janeiro, Brazil, 2006 to 2009

Middle School Teacher and Pedagogical Project Coordinator, 90+ students per year

Coordinated the engineering, design, and technology curriculum for a project-based learning school. Managed an interdisciplinary team of 10 to 15 undergraduate interns. Designed and taught experiential learning modules integrating engineering principles and design for students aged 11 to 16 through boat design and construction.

COURSES PREPARED TO TEACH

Ready to teach immediately. Operations Management · Supply Chain Management · Supply Chain Analytics · Operations Research and Deterministic Optimization · Prescriptive Analytics · Introduction to Data Analysis · Lean Operations and Process Improvement · Quality Management · Engineering Economy and Engineering Economics and Finance · Project Management

Ready with one semester of preparation. Predictive Analytics · Data Visualization · Simulation Modeling · Business Analytics · Decision and Risk Analysis · Sustainable and Responsible Supply Chains · Python for Business Analytics

Ready to develop. Operations Management and Social Impact · Sports and Performance Analytics

CURRICULUM DEVELOPMENT

- Course design and instructional materials for graduate level course, OR 6205: *Deterministic Operations Research*, Northeastern University, Spring 2026. New materials, in-class active learning exercises, assessments, and negotiated syllabus components.
- Course design, *System Engineering in Public Programs*, post-secondary teaching certification capstone, Northeastern University, 2026.
- Complete redesign of teaching materials (with Prof. McManus and Dr. Ali), IE 4516: *Quality Assurance*, Northeastern University, 2023. Restructured around experimental, simulation-based, and project-based learning; taught several resulting modules.
- Video tutorial series for Operations Research during the remote-teaching transition, UTRGV, October 2020 to April 2021 (<https://youtu.be/GOG1Yf72Pow> · <https://youtu.be/gbPII.9ZMawA> · <https://youtu.be/ISACgCrVY8w> · <https://youtu.be/rQ6V6cQWGy8>).
- Engineering, design, and technology curriculum, project-based middle school, Universidade Federal do Rio de Janeiro, 2006 to 2009.

TEACHING DEVELOPMENT / PEDAGOGICAL TRAINING

Post-secondary Teaching Certification, Northeastern University, 2025 to 2026

Teaching Fellowship (mentored teaching, Deterministic Operations Research), Northeastern University, Fall 2025 to Spring 2026

Future Business Analytics Professors Workshop, Tippie College of Business, University of Iowa, 2026

Teaching workshop, INFORMS Committee on Teaching and Learning, 2025

ADVISING AND MENTORING

- Mentoring a master's student in an independent research project (Fall 2026, in progress).
- Mentored five graduate student projects (2 doctoral, 3 master's, drawn from several engineering disciplines) in *OR 6205: Deterministic Operations Research*, Northeastern University, Spring 2026. Project topics included trip scheduling and ranking.
- Led and mentored a five-member undergraduate research team as lead researcher, STEM identity narrative inquiry study, University of Texas Rio Grande Valley, 2020 to 2021; team members went on to graduate school and independent research projects.
- Lead teaching assistant, *OR 6205: Deterministic Operations Research*.

SERVICE

Professional

Session Chair, *Cooperation and Partnership in Public Services*, INFORMS Annual Meeting, 2025

Session Chair, *Border Security*, Decision Sciences Institute Annual Meeting, 2020

University and Departmental

Vice President and Treasurer, INFORMS Student Chapter, Northeastern University, 2022 to 2023, 2025 to present
Instructor and Organizer, Notetaking Workshop for graduate students, INFORMS Student Chapter, Northeastern University, 2024

Secretary, Women in STEM Programs, University of Texas Rio Grande Valley, 2020 to 2021

LEAP Educator, Center for Diversity and Inclusion, University of Texas Rio Grande Valley, 2019

Community

Board Member, Personnel Committee and Nominating Committee, Beacon Hill Friends House, Boston, MA, 2021 to present

Volunteer Art Teacher, AIESEC Volunteer Program, Mendoza, Argentina, 2014

PROFESSIONAL EXPERIENCE

COMLURB (municipal sanitation company), Rio de Janeiro, Brazil, 2014 to 2016

Financial Analyst

Secured 100% budget approval at annual town hall meetings by presenting complex financial analysis to non-technical stakeholders. Reduced contract and inventory costs by 30% through auditing and a revised inventory policy. Built analytical frameworks that informed executive strategic planning.

MRS Global (relocation services), Rio de Janeiro, Brazil, 2012 to 2013

Financial Administrative Assistant

Designed automated validation protocols for financial control databases, reducing error-correction workload by 60%. Streamlined financial workflows across Excel, TOTVS ERP, and Access.

Nano Presentes (design manufacturing and retail), Rio de Janeiro, Brazil, 2011 to 2012

Owner and Designer

Founded and ran a small design business; secured national blog and press coverage.

Sulamerica Seguros S.A. (insurance), Rio de Janeiro, Brazil, 2010 to 2012

Customer Service Representative

Handled vehicle accident claims and negotiated emergency service provision.

Polo Nautico, Universidade Federal do Rio de Janeiro, Rio de Janeiro, Brazil, and Leeuwarden, Netherlands, 2008

Team Leader, Naval Design and Engineering Team

Led the design and construction of a solar-powered boat; secured university and industry sponsorship; earned national and international press coverage.

TECHNICAL SKILLS

Programming and analysis: Python (6+ years: Pandas, NumPy and SciPy, scikit-learn, NetworkX), R

Optimization and modeling: Gurobi, AMPL, LINDO

Reproducibility and tools: Git and GitHub, LaTeX, Overleaf, Obsidian

Teaching: backward design, active learning, project-based learning, transparent assessment, inclusive pedagogy, curriculum development, online and hybrid course delivery

LANGUAGES

Portuguese (native) · English (fluent) · Spanish (conversational)

PROFESSIONAL MEMBERSHIPS

Institute for Operations Research and the Management Sciences · Decision Sciences Institute · Tau Beta Pi

REFERENCES

Kayse Lee Maass, Ph.D.. Associate Professor and Director IE/OR Graduate Programs, Department of Mechanical and Industrial Engineering, Northeastern University · k.maass@northeastern.edu · +1 (617) 373-3629 (*dissertation advisor*)

Hiram Moya, Ph.D.. Associate Professor and Graduate Program Coordinator, Department of Manufacturing and Industrial Engineering, University of Texas Rio Grande Valley · hiram.moya@utrgv.edu · +1 (956) 665-7073 (*M.S. thesis advisor*)

Hugh McManus, Ph.D.. Associate Teaching Professor, Department of Mechanical and Industrial Engineering, Northeastern University · h.mcmanus@northeastern.edu · +1 (617) 893-7067 (*teaching reference*)